Assignment 4 Report

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Introduction

For assignment four, we were provided with a dataset with 14 features to make a random tree based classifier to identify the positive class in the Alzheimer's diagnosis. For ease for notation, we'll refer to this 14 features as x_i with $i = 0, 1, 2, ..., 13$. Notably, we are counting from zero (0) as python indexes the features.

Question 1: Decision Tree classifier

A single decision tree classifier was trained using the provided training dataset, with 202 members of the negative class, and 202 members of the positive class. For the training procedure, we use the *scikit-learn DecisionTreeClassifier* with the default settings, and seeded random state. Using cross validation for model selection with 10 folds, we selected the best possible leaf criterion from three possible choices: *gini*, *entropy* and *log loss*. The candidate models were evaluated based on balanced accuracy, and they all achieved similar results, with the *gini* feature criterion scoring 0.5296 in balanced accuracy, and the two other feature criterion achieving 0.5345, outperforming gini. Entropy and Log Loss achieving identical results makes sense, since the two criterion are identical except for a multiplicative constant.

The best performing criterion chosen for retraining was **entropy**. The final model was assessed using the test dataset provided, with 361 members of the negative class and 194 members of the positive class. The final assessment results are shown in table 1. Notably, the balanced accuracy on the independent test set was BETTER than expected given the cross validation evaluation, suggesting that training on additional data was very beneficial for this model.

Question 2: Final Decision Tree

The final decision tree from Question 1 was plotted in figure 2, with a zoom on the root provided in figure 1. As we can see from the first "split", the most important feature to differentiate the two classes is the third featured (indexed as x_2 when counting from zero).

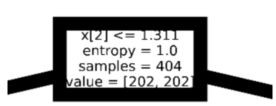

Figure 1: Zoom on the root of the tree

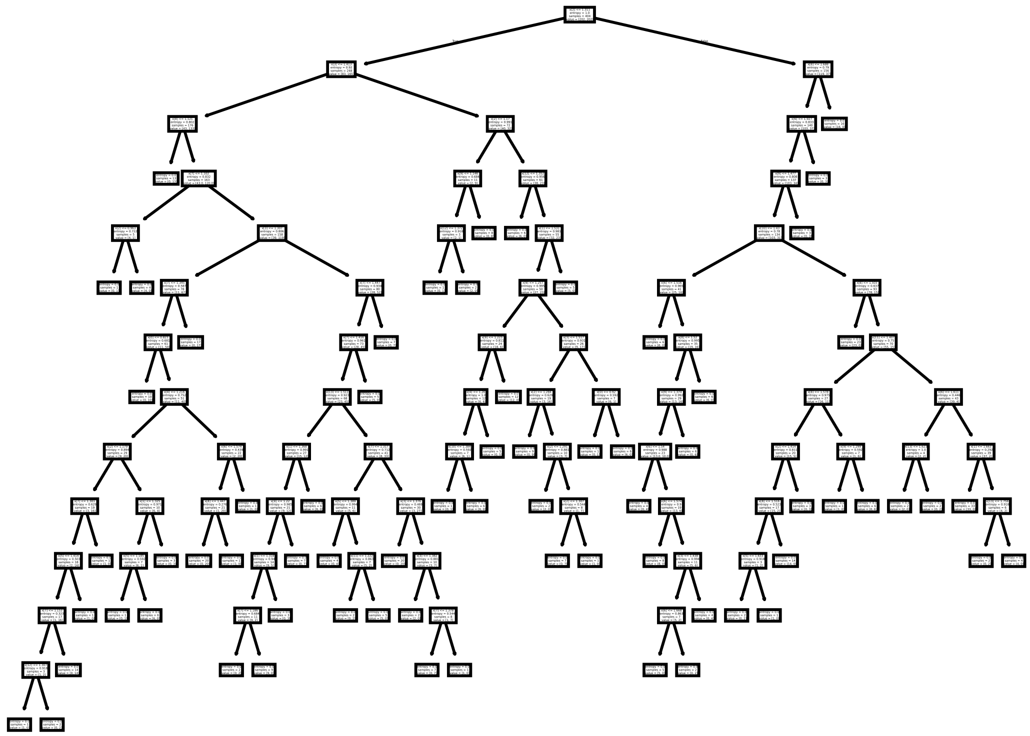

Figure 2: Final decision tree for a Decision Tree Classifier with entropy as the feature criteria

Question 3: Random Forest Classifier

For question 3, we now trained a Random Forest Classifier instead of a single decision tree. The training an testing process was identical to question 1, where we evaluated 3 possible feature criteria with a cross validation approach. For the random forest, the best feature criteria was **gini**, achieving a balanced accuracy of 0.6826 in the cross validation, versus a balanced accuracy of 0.6782 for both *log loss* and *entropy*.

The final model was retraining on the full training dataset using gini as a feature criteria, and assessed on the testing dataset provided, the results are presented in table 1. Notably, the Random Forest Classifier outperformed the decision tree in all metrics except Specificity, that is the ability to identify true negative events in the dataset.

The overall improvement in performance was expected, as the main drawback of the single decision tree is that it's prone to over fitting due to high variance, it "follows" the noise as well as the data. The random forest classifier deploys techniques like bagging and boosting to produce multiple trees with different leafs and features taken into account, and then averages their prediction. This process significantly reduces variance, making it so the classifier classifies the data, and not the noise.

Err	**Decision Tree**	**Random Forest**
Accuracy	0.5856	**0.6450**
Balanced Accuracy	0.5920	**0.6771**
Sensitivity	0.6134	**0.7835**
Specificity	0.5706	0.5706
Precision	0.4343	**0.4951**
Recall	0.6134	**0.7835**

Table 1: Model assessment results for the Decision Tree and the Random Forest Classifier, based on testing data with 361 members of the negative and 194 members of the positive class.